OSCAR CHENG

SOFTWARE ENGINEER | SEEKING INTERNSHIP

CONTACT

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Personal Website
HackMD (Tech Notes)

EDUCATION

B.S. Computer Science

National Changhua University of Education (NCUE)

September 2023 – June 2027 (Expected)

Changhua, Taiwan

SKILLS

Languages

Python, C++ (basics), JavaScript, HTML/CSS

Tools & Workflow

Git / Github, Linux, VS code

Familiar With

LLM API integration, REST APIs, MySQL basics

LANGUAGES

Mandarin Chinese — Native
English — Intermediate

SUMMARY

Computer Science undergraduate at NCUE interested in AI systems, LLM applications, and full-stack development. Currently building hands-on projects such as Market Agent, a personal market research assistant for organizing stock-related data into structured reports, and a RAG-based document analysis system. Experienced in FastAPI, vector search, API integration, and deployment, with a focus on building practical AI systems that are structured, evaluable, and maintainable.

PROJECTS

Market Agent — Personal Market Research Assistant

Python, FastAPI, Supabase, OpenRouter, yfinance, scikit-learn • 06 2026 ~ 07 2026
Repo: https://github.com/oscar940327/market-agent

- Built a personal stock research assistant that turns natural-language market questions into structured research reports covering price data, technical indicators, fundamentals, news, historical strategy evidence, and ML reference signals.
- Set up a Supabase + GitHub Actions data pipeline to update prices, technical indicators, market regimes, news, ML predictions, outcomes, and freshness checks.
- Added ML reference signals for 5 / 10 / 20-day upside probability, large-drop risk, historical return ranges, and experimental return estimates, while keeping them as research-only signals.
- Designed evidence-quality and freshness checks so each report can show how reliable the underlying data and signals are.
- Built a frontend dashboard for viewing research signals, valuation, technical timing, news impact, price plans, ML references, and structured data in one workflow.

RAG Financial Report Analysis

Python, Qdrant, Cross-Encoder, OpenAI API, Streamlit • Fall 2025

- Built a two-stage Retrieval-Augmented Generation pipeline to parse and analyze 100+ page financial reports (Tesla, Palantir), overcoming semantic retrieval ceilings in dense documents.
- Integrated a Cross-Encoder reranking engine (sentence-transformers) and an automated Standalone Query Generator to handle multi-turn context and cross-lingual queries.
- Applied anti-hallucination mechanisms through closed-book prompting, ensuring all generated analyses are grounded in retrieved data.